Project Plan Form

1. Project Title

An Implementation of the “Racetrack” Game

2. Provide a description of the project

The “Racetrack” game is a turn-based pencil-and-paper game where players must race around a track using vectors on a grid, as opposed to continuous space **[1] [2]**. Players can select a grid cell to move to, where the neighbouring cells they can select are determined by their current velocity. Selecting a cell then determines the player’s subsequent velocity. Players must carefully monitor their velocity to make their way around without crashing.

This project attempts to recreate the “Racetrack” game, primarily planned to include track building capabilities, the ability to racetracks, and AI racers of varying difficulty to compete against. These features at least provide an experience for players to challenge themselves on custom made tracks or against bot racers.

If time permits, further features may be implemented such as data management, multiplayer, and customisation to create a more game-like experience with the opportunity to interact with other players.

3. What is the main technical problem your project aims to solve?

The project aims to implement a full-fledged game incorporating the above primary features. A key challenge will be leveraging current technologies to achieve the desired functionality cohesively, which will require research and demos to assess their feasibility.

Another challenge involves implementing AI capable of driving the track along an optimal race line, while considering the shape of the track, surface materials, and any obstacles along the way. The AI will also need to have varying levels of difficulty to ensure it is not too hard to go up against, which will also be a challenge.

4. Please provide an indicative timeline for your project

The following table outlines the tasks over the project, where **Week 1** is relative to the start of the project's development. A large portion of reading has already been done prior to the anticipated start, but additional reading may be done throughout the project.

Week #	w/c	Tasks
1	20/10/25	Core UI screens (mostly placeholders)
2	27/10/25	Track representation and generating basic test track
3	03/11/25	Racing controls, basic track gameplay, and race results
4	10/11/25	Track building: point management
5	17/11/25	Extended tracks list and basic track saving
6	24/11/25	Track building: material management
7	01/12/25	Track options screen and car types
8	08/12/25	Track building: decoration management
9	15/12/25	Holidays
10	22/12/25	Holidays
11	29/12/25	Holidays
12	05/01/26	Holidays
13	12/01/26	Best-line prediction
14	19/02/26	AI racers and more race options
15	26/02/26	Turn-based multiplayer
16	02/03/26	Car customisation
17	09/03/26	Advanced data saving
18	16/03/26	Track building: drawing-based tracks
19	23/03/26	Code refinement and final testing
20	30/03/26	Code refinement and final testing
21	06/04/26	Dissertation write-up
22	13/04/26	Dissertation write-up
23	20/04/26	Dissertation write-up
24	27/04/26	Dissertation write-up
25	04/05/26	Dissertation write-up

5. Completed skills log?

Yes.

References

[1] M. A. Bekos, T. Bruckdorfer, H. Förster, M. Kaufmann, S. Poschenrieder, and T. Stüber, “Algorithms and insights for RaceTrack,” vol. 748, pp. 2–16, 2018, doi: 10.1016/j.tcs.2018.04.028.

[2] "The Racetrack Game", University of Leicester [Online]. Available at: https://campus.cs.le.ac.uk/teaching/proposals/v24167.the_racetrack_game.